

1. Administrative & Study Identification

Study Title: Study to Evaluate Efficacy and Safety of Inclisiran in Adolescents With Heterozygous Familial Hypercholesterolemia
Official Title: Two Part (Double-blind Inclisiran Versus Placebo [Year 1] Followed by Open-label Inclisiran [Year 2]) Randomized Multicenter Study to Evaluate Safety, Tolerability, and Efficacy of Inclisiran in Adolescents (12 to Less Than 18 Years) With Heterozygous Familial Hypercholesterolemia and Elevated LDL-cholesterol
Short Title/Acronym: ORION-16 Study
Protocol Number: CKJX839C12301 / 2020-002757-18
NCT Number: NCT04652726
Trial Status: COMPLETED (Study has been completed)
Sponsor/Company Name: Novartis
Investigational Product: Inclisiran (Small interfering RNA targeting PCSK9)
Trade Name: Leqvio
ATC Code: C10AX16
EMA URL: https://www.ema.europa.eu/en/documents/product-information/leqvio-epar-product-information_en.pdf
Phase of Development: PHASE3
Medical Monitor: Novartis Clinical Operations Lead

2. Study Synopsis & Rationale

2.1 Study Objective
Primary Objective: To evaluate and demonstrate the superiority of inclisiran compared to placebo in reducing low-density lipoprotein cholesterol (LDL-C) (percentage change) at Day 330 in adolescents with heterozygous familial hypercholesterolemia (HeFH).
Secondary Objectives: To evaluate the effect of inclisiran on other lipid parameters (including absolute change in LDL-C and percent change in Apo B), and to assess long-term safety, tolerability, and treatment-emergent adverse events over 720 days.

2.2 Background & Rationale
Heterozygous Familial Hypercholesterolemia (HeFH) is an inherited genetic disorder caused by inheriting one abnormal copy of a gene, most commonly the LDLR gene, responsible for clearing low-density lipoprotein (LDL) cholesterol from the blood. This genetic defect leads to a lifelong elevation of plasma LDL cholesterol levels, significantly increasing the risk of premature atherosclerotic cardiovascular disease, including heart attacks and strokes, often in early to mid-adulthood.

Currently, adolescents with HeFH have limited treatment options available to reach LDL-C treatment goals despite maximally tolerated statin therapy. Inclisiran is a small interfering RNA (siRNA) that durably lowers LDL-C. This study aims to provide necessary pediatric data on the safety, tolerability, and

efficacy of inclisiran to offer a new, infrequent (twice-yearly) dosing regimen for adolescents aged 12 to <18 years.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled (in Part 1 / Year 1); Single-arm, open-label active (in Part 2 / Year 2) **Blinding:** Double-blind (Part 1); Open-label (Part 2) **Randomization:** 2:1 (Inclisiran to Placebo) **Criteria Type / Text:** PHASE_REQUIREMENT (Trial must be in PHASE3)

3.2 Planned Enrollment & Duration Targeted Enrollment: 141 **Number of Sites:** 51 sites across 26 countries **Estimated Study Start:** 27-Jan-2021 **Estimated Primary Completion:** 09-Nov-2023

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Adolescents aged 12 to <18 years. 2. Heterozygous Familial Hypercholesterolemia (HeFH) diagnosed either by genetic testing or on phenotypic criteria. 3. Fasting LDL-C >130 mg/dL (3.4 mmol/L) at screening. 4. Fasting triglycerides <400 mg/dL (4.5 mmol/L) at screening. 5. On a maximally tolerated dose of statin (investigator's discretion) with or without other lipid-lowering therapy, stable for ≥ 30 days before screening.

4.2 Exclusion Criteria 1. Homozygous familial hypercholesterolemia (HoFH). 2. Active liver disease. 3. Secondary hypercholesterolemia (e.g., hypothyroidism or nephrotic syndrome). 4. Previous treatment with monoclonal antibodies directed towards PCSK9 within 90 days of screening. 5. Recent and/or planned use of other investigational medicinal products or devices.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: * **Part 1 (Year 1):** Inclisiran sodium 300 mg or matching placebo administered on Days 1, 90, and 270. * **Part 2 (Year 2):** Open-label inclisiran 300 mg administered to all participants (including those previously on placebo) on Days 360, 450, and 630. **Route of Administration:** Subcutaneous injection. **Duration of Treatment:** 720 days (2 years).

5.2 Concomitant & Prohibited Therapy Permitted: Stable, optimal background lipid-lowering therapy (e.g., maximally tolerated statins and/or ezetimibe). **Prohibited:** Other investigational medicinal products, devices, or PCSK9-directed monoclonal antibodies during the trial.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Days -30 to 0	Informed Consent, Medical History, Fasting Lipid Profile, Genetic/Phenotypic Assessment
Baseline	Day 1	Randomization, First Dose (Inclisiran or Placebo)
Interim Part 1	Days 90, 270	Subcutaneous injection of study drug, Safety Labs, Lipid Assessments
Primary Endpoint	Day 330	Primary Efficacy Assessment (LDL-C percentage change)
Interim Part 2	Days 360, 450, 630	Transition to Open-Label Inclisiran, Safety & Efficacy Labs
End of Study	Day 720	Final Lipid Profile, Safety Assessments, Exit Interview

7. Outcome Measures (Endpoints)

7.1 Primary Outcome 1. Percentage Change in LDL-C From Baseline to Day 330 (Part 1/Year 1)

7.2 Secondary Outcomes 1. Time-adjusted Percent Change in LDL-C From Baseline After Day 90 and up to Day 330 (Part 1/Year 1) 2. Absolute Change in LDL-C From Baseline to up Day 330 (Part 1/Year 1) 3. Percent Change in Apo B From Baseline up to Day 330 (Part 1/Year 1) 4. Long-term assessment of HDL-C, VLDL-C, triglycerides, total cholesterol, non-HDL-C, and PCSK9 levels from baseline up to Day 720.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous monitoring of treatment-emergent adverse events (e.g., injection site reactions) assessed in all patients who received at least one dose of the study drug. **Serious Adverse Events (SAEs):** Standard 24-hour reporting protocols for serious or medically significant events. **Data Monitoring Committee (DMC):** Independent data monitoring committee overseeing pediatric safety data throughout the 720-day follow-up.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) for primary efficacy assessments (all randomized patients). Safety Set for safety endpoints (all patients who received ≥ 1 dose of study drug). **Sample Size Justification:** Targeted enrollment of 141 powered to demonstrate superiority of inclisiran over placebo in percentage LDL-C reduction. **Handling of Missing Data:** Analysis of Covariance (ANCOVA) models utilized, with treatment as a fixed effect and baseline PCSK9 levels and baseline age group as covariates.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Regular onsite and remote monitoring to ensure protocol adherence and validity of central laboratory

collections. **Audit Trail:** Secure Electronic Data Capture (EDC), stringent documentation of statistical methods (SAP Amendment 4), and structured data clarification workflows.

11. Study Identifiers & Governance

Posting ID: {{276960662917842551}} Primary ID: {{CKJX839C12301}}
Registry Number: {{NCT04652726}} Sponsor/Lead Organization: {{Novartis}}
Study Title: {{Study to Evaluate Efficacy and Safety of Inclisiran in Adolescents With Heterozygous Familial Hypercholesterolemia}}
Official Regulatory Title: {{Two Part (Double-blind Inclisiran Versus Placebo [Year 1] Followed by Open-label Inclisiran [Year 2]) Randomized Multicenter Study to Evaluate Safety, Tolerability, and Efficacy of Inclisiran in Adolescents (12 to Less Than 18 Years) With Heterozygous Familial Hypercholesterolemia and Elevated LDL-cholesterol}}
Source Level: {{5}}

12. Clinical Framework (Familial Hypercholesterolemia Specific)

Primary Condition / Disease Area: {{Familial Hypercholesterolemia - Heterozygous}} Preferred Name / UMLS Name: {{Heterozygous Familial Hypercholesterolemia}} Semantic Type: {{Disease or Syndrome}} MeSH Code: {{C06.552.417}} SNOMED ID: {{238079002}}
SNOMED Hierarchy: {{398036000|Familial hypercholesterolemia (disorder) ; 238076009|Primary hypercholesterolemia (disorder) ; 13644009|Hypercholesterolemia (disorder) ; 55822004|Hyperlipidemia (disorder) ; 238037008|Disorder of lipoprotein storage and metabolism (disorder) ; 48286001|Disorder of lipoprotein AND/OR lipid metabolism (disorder) ; 75934005|Metabolic disease (disorder) ; 64572001|Disease (disorder)}}
Diagnosis Threshold: {{Fasting LDL-C > 130 mg/dL}} Medical Methodology: {{Standard of care background lipid-lowering therapy}} Optimization Target: {{LDL-C and PCSK9 reduction}}

13. Study Procedures & Methodology

Management Pathway: {{Subcutaneous injection of small interfering RNA (siRNA) targeting PCSK9}} Education Protocol (HCP): {{Site investigator training on pediatric lipid guidelines and injection administration}} Education Protocol (Patient): {{Study adherence, fasting requirements, and safety reporting instructions}} Quality

Audit Mechanism: {{Central laboratory lipid profile assessments and site safety monitoring}}

14. Observation & Follow-Up Schedule

Total Duration: {{720 Days / 2 Years}} Onsite Visit Cadence: {{Days 1, 90, 270, 330, 360, 450, 630, 720}} Remote Monitoring Frequency: {{As per trial protocol}} Checklist Review Interval: {{Routine evaluation of concomitant medications and adverse events}}

15. Sign-off & Approval

Role	Name	Signature	Date	:---	:---	:---	:---	
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]					Lead
Statistician	[Name]	_____	[DD-MMM-YYYY]					